

BAYESIAN PREDICTION IN ANALYSIS OF COVARIANCE- THE RANDOM REGRESSOR CASE

**A.J. van der
Merwe***
and
**C.A. van der
Merwe**

Department of Mathematical Statistics, University
of the Orange Free State, P.O. Box 339, Bloem-
fontein, Republic of South Africa

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Summary: In this paper Bayesian prediction under exchangeability from multivariate regression equations with random independent variables in the case of one-way analysis of covariance is considered. The work of Zellner and Soo-Bin Park (1987) is therefore extended to the proper prior case. The theory and results are applied to a real problem. Some Bayesian diagnostic checks are also given.

1. INTRODUCTION

In this paper Bayesian prediction under exchangeability from multivariate regression equations with random independent variables in the case of one-way analysis of covariance will be considered. To obtain predictions of the dependent variables, predictions of the independent variables are required. Inferences about future values of the dependent variables will therefore reflect the uncertainty about the future values of the independent variables. Feldstein (1971) considered this problem from a sampling theory

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point of view while Zellner and Soo-Bin Park (1987) applied Bayesian prediction methods to stock market data.

In the next paragraphs, the work of Zellner and Soo-Bin Park (1987) will be extended to the proper prior case. Although it is not possible to derive the exact posterior predictive density, the exact moments can be calculated. The theory and results are applied to a real problem. Numerical integration is used to approximate the unconditional posterior predictive density of the dependent variable. Some Bayesian diagnostic checks are also given.

2. THE CONDITIONAL POSTERIOR PREDICTIVE PROBABILITY DENSITY FUNCTION OF A FUTURE VECTOR

$\underline{y}_j$

Consider the following multivariate linear regression model with k random independent variables.

$$(2.1) \quad y_{ij} = \alpha_j + \beta_1(x_{ij1} - \bar{x}_{j1}) + \beta_2(x_{ij2} - \bar{x}_{j2}) + \cdots \\ + \beta_k(x_{ijk} - \bar{x}_{jk}) + \varepsilon_{ij} \\ i = 1, \dots, n; j = 1, \dots, p,$$

where

$$(2.2) \quad \varepsilon_{ij} \stackrel{\text{ind}}{\sim} N(0, \sigma_1^2)$$

and

$$(2.3) \quad \bar{x}_{j\ell} = \frac{1}{n} \sum_{i=1}^n x_{ij\ell}, \ell = 1, \dots, k.$$

Let

$$(2.4) \quad \underline{y}(np \times 1) = [y_{11} \ y_{21} \ \cdots \ y_{n1} \ y_{12} \ \cdots \ y_{n2} \ \cdots \ y_{1p} \ \cdots \ y_{np}]'$$

$$(2.5) \quad \underline{\theta}_1((p+k) \times 1) = [\alpha_1 \ \alpha_2 \ \cdots \ \alpha_p \ \beta_1 \ \beta_2 \ \cdots \ \beta_k]'$$

$$(2.6) \quad X(np \times (p+k)) = \begin{bmatrix} \underline{e} & \underline{0} & \underline{0} \cdots \underline{0} & (\underline{x}_{11} - \bar{x}_{11}\underline{e}) \cdots (\underline{x}_{1k} - \bar{x}_{1k}\underline{e}) \\ \underline{0} & \underline{e} & \underline{0} \cdots \underline{0} & (\underline{x}_{21} - \bar{x}_{21}\underline{e}) \cdots (\underline{x}_{2k} - \bar{x}_{2k}\underline{e}) \\ \underline{0} & \underline{0} & \underline{e} \cdots \underline{0} & (\underline{x}_{31} - \bar{x}_{31}\underline{e}) \cdots (\underline{x}_{3k} - \bar{x}_{3k}\underline{e}) \\ \vdots & \vdots & \vdots & \vdots \\ \underline{0} & \underline{0} & \underline{0} \cdots \underline{e} & (\underline{x}_{p1} - \bar{x}_{p1}\underline{e}) \cdots (\underline{x}_{pk} - \bar{x}_{pk}\underline{e}) \end{bmatrix}$$

where $\underline{e}(n \times 1) = [1 \ 1 \dots 1]'$; also the i -th element of $(\underline{x}_{j\ell} - \bar{x}_{j\ell}\underline{e})$ is $(x_{ij\ell} - \bar{x}_{j\ell})$.

Equation (2.1) is therefore the usual one-way analysis of covariance model with more than one independent variable. In matrix notation (2.1) can be written as:

$$(2.7) \quad \underline{y}|X, \underline{\theta}_1, \sigma_1^2 \sim N(X\underline{\theta}_1, \sigma_1^2 I_{np}).$$

where I_{np} is the $np \times np$ identity matrix.

As regards a prior probability density function for the parameters, the same improper prior

$$(2.8) \quad p(\underline{\theta}_1, \sigma_1) = p(\underline{\theta}_1)p(\sigma_1) \propto \sigma_1^{-1}$$

as given by Zellner and Soo-Bin Park (1987) will be used. This means that the parameters $\alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_k$ and $\log \sigma_1$ are independently and uniformly distributed.

Let

$$(\underline{x}_{11f} \dots \underline{x}_{1kf}, y_{1f}); \dots; (\underline{x}_{p1f} \dots \underline{x}_{pkf}, y_{pf})$$

be the future unknown values of the independent and dependent variables from each of the p groups. Using (2.1)-(2.7) it can easily be shown that the posterior predictive probability density function of a future vector of observations $\underline{y}_f = (y_{1f}, y_{2f}, \dots, y_{pf})'$ conditional on $\tilde{X}_f$ and the data is given by

$$(2.9) \quad f(\underline{y}_f | \tilde{X}_f, X, y) = \frac{\Gamma[\frac{1}{2}(p + v_1)](v_1 S_{y.x}^2)^{\frac{1}{2}v_1}}{\Gamma(\frac{1}{2}v_1)\Pi^{p/2}|G_1|^{\frac{1}{2}}[v_1 s_{y.x}^2 + (\underline{y}_f - \hat{\underline{y}}_f)'G_1^{-1}(\underline{y}_f - \hat{\underline{y}}_f)]^{\frac{1}{2}(p+v_1)}},$$

a multivariate Student t -distribution (Zellner and Soo-Bin Park (1987) page 241) with mean

$$(2.10) \quad E(\underline{y}_f | \tilde{X}_f, X, y) = \hat{\underline{y}}_f = \tilde{X}_f(X'X)^{-1}X'y$$

and covariance matrix

$$(2.11) \quad \begin{aligned} Var(\underline{y}_f | \tilde{X}_f, X, y) &= \frac{v_1 S_{y.x}^2}{v_1 - 2} \{I_p + \tilde{X}_f(X'X)^{-1}\tilde{X}_f'\} \\ &= \frac{v_1 S_{y.x}^2}{v_1 - 2} G_1 \end{aligned}$$

where I_p is the $p \times p$ identity matrix,

$$(2.12) \quad \tilde{X}_f(p \times (p+k)) = \begin{bmatrix} 1 & 0 & 0 \cdots 0 & (x_{11f} - \bar{x}_{11}) \cdots (x_{1kf} - \bar{x}_{1k}) \\ 0 & 1 & 0 \cdots 0 & (x_{21f} - \bar{x}_{21}) \cdots (x_{2kf} - \bar{x}_{2k}) \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 \cdots 1 & (x_{p1f} - \bar{x}_{p1}) \cdots (x_{pkf} - \bar{x}_{pk}) \end{bmatrix}$$

$$(2.13) \quad S_{y.x}^2 = \frac{1}{v_1} (\underline{y}'\underline{y} - \underline{y}'X(X'X)^{-1}X'\underline{y}),$$

the error variance, and

$$(2.14) \quad v_1 = np - p - k \quad (v_1 > 2).$$

Equation (2.9) results from the fact that the posterior predictive probability density of $\underline{y}_f$ given $\tilde{X}_f, \sigma_1^2$ and the data is normal with mean $\underline{\hat{y}}_f$ and covariance matrix

$$(2.15) \quad \text{Var}(\underline{y}_f | \tilde{X}_f, \sigma_1^2, X, \underline{y}) = \sigma_1^2 G_1$$

and that the posterior density of σ_1 ; i.e.

$$(2.16) \quad f(\sigma_1 | S_{y.x}^2) \propto \sigma_1^{-(v_1+1)} \exp\left\{-\frac{1}{2\sigma_1^2}(v_1 S_{y.x}^2)\right\}, \quad \sigma_1 > 0.$$

By multiplying the normal density with (2.16) and integrating over σ_1 , (2.9) follows.

As mentioned in the introductory paragraph, the main purpose of this paper is to extend the work of Zellner to the proper prior case. Therefore, in the next paragraph the special case, $k = 1$, will be considered while in the last paragraph some Bayesian diagnostic checks are given.

3. ONE INDEPENDENT VARIABLE ($k = 1$)

3.1 THE POSTERIOR PREDICTIVE PROBABILITY DENSITY FUNCTION OF $\underline{x}_f = [x_{1f} x_{2f} \cdots x_{pf}]'$

To obtain the unconditional posterior predictive probability density function of $\underline{y}_f$, the posterior predictive probability density function of $\underline{x}_f$ is

needed. $x_{1f}, x_{2f}, \dots, x_{pf}$ are future observations from each of the p groups. Therefore assume that

$$(3.1.1) \quad x_{ij} | \zeta_j, \sigma_2^2 \stackrel{\text{ind}}{\sim} N(\zeta_j, \sigma_2^2) \quad j = 1, \dots, p \quad i = 1, \dots, n$$

which means that the variances within the groups are the same. From (3.1.1) it follows that

$$(3.1.2) \quad \underline{x}_f | \theta_2, \sigma_2^2 \sim N(\theta_2, \sigma_2^2 I_p)$$

where

$$(3.1.3) \quad \underline{\theta}_2 = [\zeta_1 \zeta_2 \dots \zeta_p]'$$

and I_p is the $p \times p$ identity matrix.

Suppose further that the parameters $\zeta_1, \dots, \zeta_p$ of the independent variables across the p groups are exchangeable. This means (Lindley and Smith (1972)) that their distribution would be unaltered by any permutation of the suffixes so that, in particular, the prior opinion of ζ_5 is the same as that for ζ_2 , or any other ζ_j and similarly for pairs, triplets and so on. Based on a liberal interpretation of a mathematical result in de Finetti (1963) the exchangeability assumption implies that the prior distribution of the ζ_j can be chosen to be i.i.d. given some hyperparameters (e.g. μ and τ^2) that also have a prior distribution.

Therefore, assume a normal prior

$$(3.1.4) \quad \underline{\theta}_2 | \mu, a, \sigma_2^2 \sim N(\mu \underline{e}, \tau^2 I_p)$$

where $\tau^2 = a\sigma_2^2$ and independent of $p(\underline{\theta}_1, \sigma_1)$; where $\underline{e}(p \times 1) = [1 \ 1 \dots 1]'$ and $0 < a < \infty$.

For convenience sake, we adopt the representation $\tau^2 = a\sigma_2^2$ for the prior variance. This will facilitate the integration over σ_2 . The variable a on the other hand, will always occur in the form $b = \frac{1}{1+an}$ in the posterior density function. See for example Press and Shigemasu (1985).

The hierarchical Bayes approach assumes second stage priors on the hy-

perparameters. The difficulty of specifying second stage priors has made common the use of noninformative priors at the second stage. We shall therefore assume that

$$(3.1.5) \quad P(\mu, \sigma_2) = p(\mu)p(\sigma_2) \propto \sigma_2^{-1} \quad 0 < \sigma_2 < \infty$$

and independently distributed of $\underline{\theta}_1$, $\underline{\theta}_2$ and σ_1^2 .

A prior can be put on a or $b = \frac{1}{1+an}$. Strawderman (1971) considered the following family of priors for b

$$(3.1.6) \quad p(b) \propto b^{-\phi} \quad 0 < b < 1.$$

The case $\phi = 2$ corresponds to a uniform prior on a . Since a is not a variance $\phi = 1$ will rather be used.

$p(b) \propto b^{-1}$ then corresponds to a Jeffreys type prior structure on the within and between variance. This prior structure has been studied by Hill (1965) and used by Box and Tiao (1973 chapter 7).

Equations (3.1.1)-(3.1.6) represent the simplest hierarchical model of any practical value and were used by Lindley and Smith (1972) and Box and Tiao (1973) among others for data having the one-way analysis of variance structure. There are good reasons to believe that the feed consumption data given in Table 3.3.1 will satisfy these assumptions. For further discussions on model assumptions and diagnostic checking of these data, see paragraph 4 and Appendix A.

Using equations (3.1.2)-(3.1.5) it can easily be shown (Box and Tiao (1973) and Van der Merwe and Smit (1986) that the posterior predictive density function of $\underline{x}_f$ given the data and b is

$$(3.1.7) \quad f(\underline{x}_f | X, b) = \frac{(bS+W)^{\frac{1}{2}(p+v_2-1)} \Gamma\left(\frac{2p+v_2-1}{2}\right)}{\Gamma\left(\frac{p+v_2-1}{2}\right) (\pi)^{p/2} |G_2|^{\frac{1}{2}} \left\{ (bS+W) + (\underline{x}_f - \underline{\hat{x}}_f)' G_2^{-1} (\underline{x}_f - \underline{\hat{x}}_f) \right\}^{\frac{1}{2}(2p+v_2-1)}}$$

$$-\infty \leq \underline{x}_f \leq \infty$$

a multivariate Student t -distribution with mean

$$(3.1.8) \quad E(\underline{x}_f | X, b) = \hat{\underline{x}}_f = \bar{\underline{x}} - b(\bar{\underline{x}} - \bar{\bar{x}} \underline{e})$$

and covariance matrix

$$(3.1.9) \quad \text{Var}(\underline{x}_f | X, b) = \frac{(W + bS)G_2}{(p + v_2 - 3)}$$

where

$$(3.1.10) \quad S = \sum_{j=1}^p n(\bar{x}_j - \bar{\bar{x}})^2$$

$$(3.1.11) \quad G_2 = I_p \left(1 + \frac{1}{n}\right) - \frac{b}{n} \left(I_p - \frac{1}{p} \underline{e} \underline{e}'\right)$$

$$(3.1.12) \quad v_2 = p(n - 1)$$

$$(3.1.13) \quad W = \sum_{j=1}^p \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$$

and

$$(3.1.14) \quad \bar{\bar{x}} = \frac{1}{p} \sum_{j=1}^p \bar{x}_j.$$

To obtain the unconditional predictive density of $\underline{x}_f$, the posterior density of b must first be obtained. By integrating the joint posterior density function of all the parameters over $\underline{\theta}_2, \mu$ and σ^2 , the posterior density of b is found to be

$$(3.1.15) \quad f(b|c) = K_1 b^{\frac{1}{2}(p-1)-1} (1 + cb)^{-\frac{1}{2}(p+v_2-1)} \quad 0 < b \leq 1$$

which is a truncated scaled F-density with K_1 the normalizing constant and

$$(3.1.16) \quad c = \frac{S}{W} = \frac{n \sum_{j=1}^p (\bar{x}_j - \bar{\bar{x}})^2}{\sum_{j=1}^p \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}.$$

Equation (3.1.15) corresponds to Box and Tiao's equation (7.2.26).

For p uneven, an expression for the h -th moment of $f(b|c)$ can be obtained analytically (Van der Merwe and Smit (1986) and Box and Tiao (1973)) and for p even the posterior moments of b can be calculated via numerical integration.

The unconditional posterior predictive probability density function of $\underline{x}_f$ is therefore given by

$$(3.1.17) \quad f(\underline{x}_f|X) = \int_0^1 f(\underline{x}_f|X, b) f(b|c) db.$$

and the unconditional posterior predictive probability density function of $\underline{y}_f$ by

$$(3.1.18) \quad f(\underline{y}_f|X, \underline{y}) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(\underline{y}_f|\underline{x}_f, X, \underline{y}) f(\underline{x}_f|X) d\underline{x}_f.$$

Although these integrals can be evaluated numerically, they cannot be obtained in closed form. It is however possible to calculate the exact means and covariance matrices of (3.1.17) and (3.1.18) as well as higher moments of individual components of $\underline{x}_f|X$ and $\underline{y}_f|X, \underline{y}$.

3.2 THE MEANS AND COVARIANCE MATRICES OF $\underline{x}_f$ and $\underline{y}_f$.

In the case of one independent variable the mean and covariance matrix of the conditional posterior predictive density of $\underline{y}_f$ ((2.10) and (2.11)) can be simplified to

$$(3.2.1) \quad E(\underline{y}_f|\underline{x}_f, X, \underline{y}) = \hat{\underline{y}}_f = \bar{\underline{y}} + (\underline{x}_f - \bar{\underline{x}})\hat{\beta}$$

and

$$(3.2.2) \quad \text{Var}(\underline{y}_f | \underline{x}_f, X, \underline{y}) = G_1 \frac{v_1 S_{\underline{y}.x}^2}{v_1 - 2}$$

where

$$(3.2.3) \quad \underline{\bar{y}} = (\bar{y}_1 \cdots \bar{y}_p)'$$

$$(3.2.4) \quad \bar{y}_j = \frac{1}{n} \sum_{i=1}^n y_{ij} \quad j = 1, \dots, p$$

$$(3.2.5) \quad \hat{\beta} = \frac{W_{xy}}{W}$$

$$(3.2.6) \quad W = W_{xx} = \sum_{j=1}^p \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$$

$$(3.2.7) \quad W_{xy} = \sum_{j=1}^p \sum_{i=1}^n (x_{ij} - \bar{x}_j)(y_{ij} - \bar{y}_j)$$

$$(3.2.8) \quad G_1 = (1 + \frac{1}{n})I_p + \frac{1}{W}(\underline{x}_f - \underline{\bar{x}})(\underline{x}_f - \underline{\bar{x}})'$$

$$(3.2.9) \quad \text{and } v_1 = np - p - 1.$$

The following theorem can now be stated:

Theorem 3.2.1

The means and covariance matrices of the unconditional posterior predictive densities of $\underline{x}_f$ and $\underline{y}_f$ are given by

$$(3.2.10) \quad E(\underline{x}_f | X) = \underline{\bar{x}} - E(b|c)(\underline{\bar{x}} - \underline{\bar{x}} \underline{\bar{e}})$$

$$(3.2.11) \quad E(\underline{y}_f | \underline{y}, X) = \underline{\bar{y}} - E(b|c)(\underline{\bar{x}} - \underline{\bar{x}} \underline{\bar{e}}) \hat{\beta}$$

$$(3.2.12) \quad \begin{aligned} Var(\underline{x}_f | X) = & \left(\frac{1}{p+v_2-3} \right) \left\{ W(1 + \frac{1}{n}) I_p \right. \\ & + E(b|c) [S(1 + \frac{1}{n}) I_p - \frac{1}{n} W(I_p - \frac{1}{p} \underline{e} \underline{e}')] \\ & \left. - E(b^2|c) \frac{1}{n} S(I_p - \frac{1}{p} \underline{e} \underline{e}') \right\} + (\underline{\bar{x}} - \underline{\bar{x}} \underline{\bar{e}}) (\underline{\bar{x}} - \underline{\bar{x}} \underline{\bar{e}})' Var(b|c) \end{aligned}$$

$$(3.2.13)$$

$$\begin{aligned} Var(\underline{y}_f | \underline{y}, X) = & \left\{ \frac{\hat{\beta}^2}{p+v_2-3} W + \left(\frac{v_1 S_{\underline{y}, \underline{x}}^2}{v_1-2} \right) \left(\frac{p+v_2-2}{p+v_2-3} \right) \right\} (1 + \frac{1}{n}) I_p \\ & + E(b|c) \left\{ \left(\hat{\beta}^2 + \frac{v_1 S_{\underline{y}, \underline{x}}^2}{(v_1-2)W} \right) \left(\frac{1}{p+v_2-3} \right) \left(S(1 + \frac{1}{n}) I_p - \frac{W}{n} (I_p - \frac{1}{p} \underline{e} \underline{e}') \right) \right\} \\ & + E(b^2|c) \left\{ \left(\frac{v_1 S_{\underline{y}, \underline{x}}^2}{v_1-2} \right) \frac{1}{W} (\underline{\bar{x}} - \underline{\bar{x}} \underline{\bar{e}}) (\underline{\bar{x}} - \underline{\bar{x}} \underline{\bar{e}})' - \frac{S}{n(p+v_2-3)} \left(\hat{\beta}^2 + \frac{v_1 S_{\underline{y}, \underline{x}}^2}{(v_1-2)W} \right) \right. \\ & \left. \left(I_p - \frac{1}{p} \underline{e} \underline{e}' \right) \right\} + Var(b|c) \hat{\beta}^2 (\underline{\bar{x}} - \underline{\bar{x}} \underline{\bar{e}}) (\underline{\bar{x}} - \underline{\bar{x}} \underline{\bar{e}})'. \end{aligned}$$

The proofs of (3.2.10) - (3.2.13) as well as expressions for $E(y_{fj}^3 | X, \underline{y})$ and $E(y_{fj}^4 | X, \underline{y})$ are given in Van der Merwe and Van der Merwe (1990).

It is interesting to note that for the improper prior case (Zellner and Soo-Bin Park (1987)) the mean of the predictive density, $E(\underline{y}_f | X) = \underline{\bar{y}}$, while for the proper prior case the X -data are also included in the predictive mean, $E(\underline{y}_f | \underline{y}, X) = \underline{\bar{y}} - E(b|c)(\underline{\bar{x}} - \underline{\bar{x}} \underline{\bar{e}}) \hat{\beta}$.

The theory and results obtained in this and previous paragraphs as well as those given in the Appendix will now be applied to a real problem.

3.3 GAIN IN WEIGHT AND FEED CONSUMPTION OF PIGS IN COMPARATIVE FEEDING TRIALS.

Crampton and Hopkins (1934) studied the effects of feed consumption on the weight gaining ability of pigs. The feeding trials were conducted at five different experimental stations. Ten pigs were assigned completely at random to each trial and all pigs were fed the same type of feed. The data

in table 3.3.1 are the gain in weight (Y) and the amount of feed consumed (X) over a period of time. Both measurements are in pounds.

One of the main reasons for conducting this experiment was to estimate the expected feed intake and gain in weight of pigs in future experiments at different experimental stations and to obtain confidence intervals for these parameter values.

As mentioned before and as will be shown in paragraph 4 and Appendix A, there is good reason to believe that these data satisfy the model assumptions for the Bayesian summary presented in sections 3.1 and 3.2.

From table 3.3.1 it therefore follows that $\hat{\beta} = 0.2233$, $c = 0.2256$ and by using simple integration, $E(b|c) = 0.3638$ and $E(b^2|c) = 0.1841$. By using equations (3.2.10) and (3.2.11) the means of the posterior predictive density functions (hierarchical Bayes estimators) $E(x_{fj}|X)$ and $E(y_{fj}|X, \underline{y})$ were calculated. These as well as the usual estimators ($\bar{x}_j$ and $\bar{y}_j$) are given in the Table 3.3.2.

Table 3.3.2 shows the shrinkage effect of the hierarchical Bayes estimates. The shrinkage effect in the case of feed consumption is much more severe than in the case of gain in weight. The reason for this is that $\hat{\beta} = 0.223$ is quite small. (The sample correlation coefficient $r = 0.7845$ is highly significant: $p < 0.01$).

By using equations (3.2.12) and (3.2.13) the standard deviations of the posterior predictive density functions were calculated and are given in Table 3.3.3.

Table 3.3.1

Gain in Weight and Feed Consumption of Pigs in Comparative Feeding Trials (in pounds)

Trial 1		Trial 2		Trial 3		Trial 4		Trial 5	
Feed eaten	Gain	Feed eaten	Gain	Feed eaten	Gain	Feed eaten	Gain	Feed eaten	Gain
X	Y	X	Y	X	Y	X	Y	X	Y
674	165	699	168	708	164	716	185	831	201
628	156	626	180	614	156	769	195	754	189
661	159	668	180	733	189	733	186	722	173
694	167	668	166	663	138	742	201	728	193
713	170	707	170	607	153	624	165	646	164
585	146	651	161	745	190	710	175	678	160
575	130	672	171	637	160	742	187	763	200
638	151	660	169	662	172	648	177	625	142
632	164	769	179	609	142	628	166	710	184
637	158	666	191	596	155	601	165	651	149
Mean									
643.7	156.6	678.6	173.5	657.4	161.9	691.3	180.2	710.8	175.5

Table 3.3.2

Hierarchical Bayes and Usual Estimates for Gain in Weight and Feed Consumption of Pigs in Comparative Feeding Trials

	Trial	1	2	3	4	5
Feed Consumption (pounds)	$\bar{x}_j$	643.7	678.6	657.4	691.3	710.8
	$E(x_{jj} X)$	655.6	677.8	664.3	685.9	698.3
Gain in weight (pounds)	$\bar{y}_j$	156.6	173.5	161.9	180.2	175.5
	$E(y_{jj} X, y)$	159.3	173.3	163.2	179.9	172.7

Table 3.3.3

Standard Deviations of the Posterior Predictive Density Functions for Feed Consumption (X) and Gain in Weight (Y) of Pigs in Comparative Feeding Trials

Trial	1	2	3	4	5
Std($x_{fj} X$) (pounds)	56.2	55.7	55.9	55.8	56.3
Std($y_{fj} X, \underline{y}$) (pounds)	16.2	16.1	16.2	16.1	16.2

where

$$Std(x_{fj}|X) = \sqrt{Var(x_{fj}|X)} \text{ and } Std(y_{fj}|X, \underline{y}) = \sqrt{Var(y_{fj}|X, \underline{y})}.$$

To assess the usefulness of the Bayes procedure, a construction sample was obtained by drawing 5 pigs at random from each trial. The data are given in Table 3.3.4.

Table 3.3.4

Gain in Weight and Feed Consumption of Pigs in Comparative Feeding Trials - Original Experiment.

Trial 1		Trial 2		Trial 3		Trial 4		Trial 5	
X	Y	X	Y	X	Y	X	Y	X	Y
628	156	626	180	662	172	716	185	710	184
661	159	699	168	663	138	710	175	646	164
585	146	707	170	708	164	624	165	831	201
638	151	672	171	637	160	628	166	754	189
674	165	668	166	614	156	769	195	763	200
Mean									
637.2	155.4	674.4	171.0	656.8	158.0	689.4	177.2	740.8	187.6

These data, which we shall call the "original" experiment, will now be used to estimate the expected feed intake and gain in weight of pigs in future experiments. The remaining data of Table 3.3.1 can be considered as an example of such a "future" experiment. The mean feed intake and gain in

weight for these data as well as the Bayes and usual estimates obtained from Table 3.3.4 are given in Table 3.3.5.

Table 3.3.5

Hierarchical Bayes and Usual Estimates as well as the Average Feed Consumption and Gain in Weight of Pigs in a Future Experiment.

		Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
Feed Consumption	$\bar{x}_j$	637.2	674.4	656.8	689.4	740.8
	$E(x_{fj} X)$	650.4	676.0	663.9	686.4	721.8
	Average Future Experiment	650.2	682.8	658.0	693.2	680.8
Gain in Weight	$\bar{y}_j$	155.4	171	158	177.2	187.6
	$E(y_{fj} X, \underline{y})$	157.5	171.2	159.1	176.7	184.3
	Average Future Experiment	157.8	176.0	165.8	183.2	163.4

From the table it is clear that, except for trials 3 and 4 in the case of feed consumption and trial 4 for gain in weight, the means of the posterior predictive density functions ($E(x_{fj}|X)$ and $E(y_{fj}|X, \underline{y})$) are better estimates of the true values than the usual estimates ($\bar{x}_j$ and $\bar{y}_j$). The reason for the serious over-prediction in trial 5 is that (by chance) most of the "large" values in column 5, Table 3.3.1, were included in the "original" experiment and the small ones in the "future" experiment.

Table 3.3.6 compares the estimated risks of the two procedures.

$$\begin{aligned}
 R(\bar{x}) &= \frac{1}{25} \{ (674 - 637.2)^2 + \cdots + (632 - 637.2)^2 + (668 - 674.4)^2 \\
 &\quad + \cdots + (651 - 740.8)^2 \} \\
 &= 3484.7,
 \end{aligned}$$

where the first values in the brackets are the feed consumption data in the future experiment and the second values the arithmetic means of the original experiments.

Table 3.3.6

A Comparison of the Estimated Risks for the Bayes Procedures ($R(\underline{x}_f)$ and $R(\underline{y}_f)$) and the Usual Procedures ($R(\bar{x})$ and $R(\bar{y})$)

Feed Consumption		Gain in Weight	
Usual Procedure	Bayes Procedure	Usual Procedure	Bayes Procedure
$R(\bar{x})$	$R(\underline{x}_f)$	$R(\bar{y})$	$R(\underline{y}_f)$
3484.7	3026.5	381.6	347.2

Also

$$\begin{aligned}
 R(\underline{x}_f) &= \frac{1}{25} \{ (674 - 650.4)^2 + \dots + (632 - 650.4)^2 + (668 - 676.1)^2 \\
 &\quad + \dots + (651 - 721.8)^2 \} \\
 &= 3026.5,
 \end{aligned}$$

where the second values in brackets are the posterior predictive means of the "original" experiment. $R(\bar{y})$ and $R(\underline{y}_f)$ are calculated in the same way.

From the table it is clear that the improvement of the Bayes estimator over the usual estimator in the case of Feed Consumption = $\frac{3484.7 - 3026.5}{3484.7} \times 100\% = 13.1\%$ while the percentage improvement in the case of Gain in Weight = $\frac{381.6 - 347.2}{381.6} \times \frac{100}{1}\% = 9.0\%$.

4. BAYESIAN SUMMARY AND JUSTIFICATION OF MODEL ASSUMPTIONS

The Bayesian summary presented in sections (3.1) and (3.2) is based on several model assumptions.

- (i) Given X — the normality of the Y -data and a common variance, σ_1^2 — Assumption (2.1).
- (ii) Given $\underline{\theta}_2$ — the normality of the X -data and a constant common variance, σ_2^2 — Assumption (3.1.1).
- (iii) The exchangeability (i.e. i.i.d. nature) of the prior distribution of each ζ_j

(iv) The normality of the prior distribution of each ζ_j given μ and τ^2 .

(v) Noninformative priors at the second stage - Assumption (3.1.5).

The first and second modelling assumptions can easily be verified. The usual tests on the errors, for example, D'Agostino's D for normality, Bartlett's test for a constant common variance, Cook's distance measure for influential observations, Grubbs and Beck's test for outliers, the Durbin-Watson and runs test for autocorrelation etc., show no violations of model assumptions.

In addition to these tests, a Bayesian approach to detecting outliers and influential observations in a linear model (assumption 2.1) is given in Appendix A, paragraph A.1. Outliers can be detected by examining the posterior distribution of the realized regression error terms. For further details, see Appendix A as well as Zellner (1975) and Chaloner and Brant (1988).

The third modelling assumption deserves commentary. The real-world interpretation of this mathematical assumption of exchangeability of the ζ_j is that, before seeing the data of the experiment, there is no desire to include in the model features such as

(a) the mean feed consumption in trial 1 is probably larger than in trial 2 or

(b) the feed consumption means in trials 1 and 2 are, for example, more similar than those in trials 1 and 3. In other words, the exchangeability assumption means that we will let the data tell us about the relative ordering and similarity of the feed consumptions means. Such an a priori stance seems reasonable for the feed consumption data given in Table 3.3.1. Practically model (3.1.4) is a device to obtain answers between the five independent estimates and the one common estimate and experience in a variety of contexts (see for example Rubin

(1981)) suggests that such intermediate answers are superior to either extreme.

The fourth and fifth modelling assumptions are harder to justify a priori. One possible way of doing this is to analyse the posterior distribution of the realized error terms under these models. These calculations are given in Appendix A paragraph A.2.

A further method for checking the adequacy of all our models is according to Rubin (1981) called "phenomenological Bayesian monitoring." The idea is to monitor the entire model being used to make sure that it could generate the observed data; the spirit is similar to that of significance testing but is Bayesian in that all inferences are conditional on the observed data. Specifically, the method is as follows. The posterior predictive density functions (Equations (3.1.17) and (3.1.18)) i.e. the data that can be expected in further replications of the trials are compared to

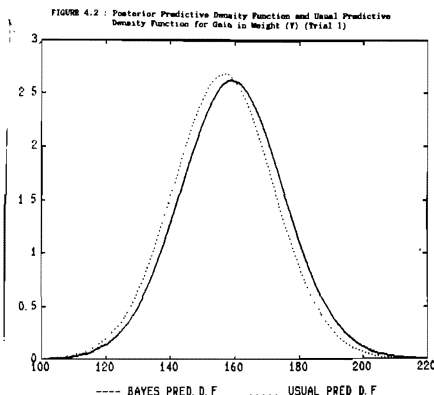
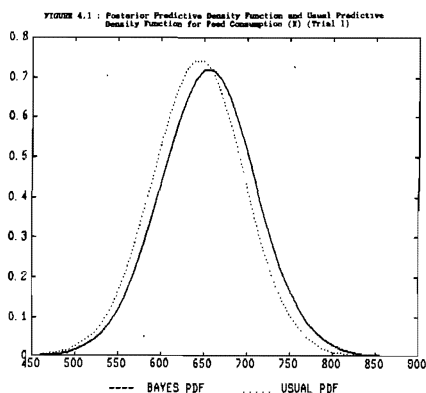
- (a) the actual data in the original trials and
- (b) scientific judgement about plausible values of such data.

If the model is an appropriate one, the hypothetical data generated under the model will not be in conflict with the actual data but will be typical of them. According to Rubin the word "phenomenological" is used to describe this monitoring of Bayesian models because the focus when checking the model is on phenomena, not unobservable parameters. From equations (3.1.17) and (3.1.18) the posterior predictive density functions for feed consumption (X) and gain in weight (Y) were obtained by integrating numerically. The density functions for Trial 1, as well as the usual predictive density functions $t_{v_2}(\bar{x}_1, \hat{Var}(x_{f1}))$ and $t_{v_2}(\bar{y}_1, \hat{Var}(y_{f2}))$ are shown in figures 4.1 and 4.2.

Now $t_{v_2}(\bar{x}_1, \hat{Var}(x_{f1}))$ is the univariate student-t p.d.f. with mean $\bar{x}_1$, $\widehat{Var}(x_{f1}) = (1 + \frac{1}{n})\frac{1}{v_2}W$ and v_2 degrees of freedom. Similarly $t_{v_2}(\bar{y}_1, \widehat{Var}(y_{f1}))$; where $\widehat{Var}(y_{f1}) = (1 + \frac{1}{n})\frac{1}{v_2}W_{yy}$.

From these figures and tables (3.3.2) and (3.3.3) it is clear that the values generated by the Bayesian model are in agreement with the observed values

of the data (Table 3.3.1) but in addition the model-generated values are all plausible outcomes of experiments on feed consumption.



A possible range of outcomes for feed consumption under the Bayesian model is

$$(E(x_{f1}|X) - 2Std(x_{f1}|X)) - (E(x_{f5}|X) + 2Std(x_{f5}|X)) \\ = 543.2 - 810.9$$

and for gain in weight

$$126.9 - 205.1.$$

Both these ranges are consistent with the ranges of the observed values. Hence, the model seems to generate data that are consistent with observed values and scientific judgment, and so conclusions based on the Bayesian model will be respectable.

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APPENDIX A

A.1 A BAYESIAN APPROACH TO RESIDUAL ANALYSIS AND OUTLIER DETECTION (Assumption 2.1)

Consider again paragraph 2 equations (2.1) and (2.7). A given sample of observations

$$\underline{y}(np \times 1) = [y_{11}y_{21} \cdots y_{n1}y_{12} \cdots y_{n2} \cdots y_{1p} \cdots y_{np}]'$$

is a realization of the process (2.7) and satisfies

$$(A.1.1) \quad \underline{y} = X\underline{\theta}_1 + \underline{u}$$

where $\underline{u}$ is an $nx1$ vector whose elements are realizations of the process generating $\underline{\varepsilon}$. From the posterior distribution of $\underline{\theta}_1$ it follows (Zellner (1975)) that the posterior distribution of $\underline{u}$ given $\underline{y}$ and X is a multivariate vector student t-distribution with mean

$$(A.1.2) \quad E(\underline{u}|X, \underline{y}) = \underline{y} - X\hat{\underline{\theta}}_1 = \hat{\underline{u}}$$

and covariance matrix

$$(A.1.3) \quad Var(\underline{u}|X, \underline{y}) = \frac{X(X'X)^{-1}X'v_1S_{\underline{y}.x}^2}{(v_1 - 2)}$$

where

$$(A.1.4) \quad \hat{\underline{\theta}}_1 = (X'X)^{-1}(X'\underline{y})$$

and $S_{\underline{y}.x}^2$ is defined in equation (2.13)

Since $H = X(X'X)^{-1}X'$ is an $np \times np$ idempotent matrix with rank p , the posterior covariance matrix for $\underline{u}$ is singular. The usual residual $\hat{u}_{ij}$ ($i = 1, \dots, n; j = 1, \dots, p$), can therefore be viewed as the posterior mean of the actually occurring random error u_{ij} . A $(1-\alpha)$ highest posterior density (hpd) interval for u_{ij} is given by

$$(A.1.5) \quad \hat{u}_{ij} \pm t(\alpha/2, np - p - 1) s_{y.x} \sqrt{h_{ij,ij}}$$

where $t(\alpha/2, np - p - 1)$ is the upper $\alpha/2$ percentage point for a t -distribution with $np - p - 1$ degrees of freedom and $h_{ij,ij}$ is the ij, ij element of H . ($i = 1, \dots, n; j = 1, \dots, p$)

The probability $P(|u_{ij}| > \tilde{k}\sigma_1 | y)$ can be defined (Chaloner and Brant (1988)) as the probability that the ij -th observation will be an outlier. These probabilities can easily be computed using numerical integration. Denote $\Phi(z)$ to be the probability that a standard normal, $N(0, 1)$, random variable is less than z . Further denote

$$z_1 = (\tilde{k} - \sigma_1^{-1} \hat{u}_{ij}) h_{ij,ij}^{-\frac{1}{2}} \quad \text{and} \quad z_2 = (-\tilde{k} - \sigma_1^{-1} \hat{u}_{ij}) h_{ij,ij}^{-\frac{1}{2}}$$

then we have

$$(A.1.6) \quad \begin{aligned} P(|u_{ij}| > \tilde{k}\sigma_1 | y) &= \int_0^\infty P(|u_{ij}| > \tilde{k}\sigma_1 | y, \sigma_1) p(\sigma_1 | y) d\sigma_1 \\ &= \int_0^\infty (1 - \Phi(z_1) + \Phi(z_2)) p(\sigma_1 | y) d\sigma_1 \end{aligned}$$

where $p(\sigma_1 | y)$ is defined in equation (2.16).

As these diagnostic measures are probabilities, they are easy to interpret. Points with high probabilities of being an outlier will have a large $|u_{ij}|$, a large $h_{ij,ij}$ or both. When $|\hat{u}_{ij}|$ is large this suggests that $|u_{ij}|$ is large. When $h_{ij,ij}$ is large, there is uncertainty about u_{ij} as reflected in the posterior variance. The quantity $h_{ij,ij}$ is often referred to as leverage, as points with large $h_{ij,ij}$ are potentially highly influential. For the "gain in weight" and "feed consumption" data (Table 3.3.1) the posterior probabilities (for $\tilde{k} = 3$) as well as the diagonal elements of the $H = X(X'X)^{-1}X'$ matrix are given in Table A.1.1.

The value of u_{43} is reasonably large. The standardized residual

$$\frac{\hat{u}_{43}}{S_{y.x}(1 - h_{43,43})^{-\frac{1}{2}}} = -2.53.$$

Table A.1.1

The Posterior Probabilities that Outliers will occur in the Y-Data (Gain in Weight Data) as well as the Diagonal Elements of $H = X(X'X)^{-1}X'$

Trial 1		Trial 2		Trial 3		Trial 4		Trial 5	
$h_{ij,ij}$	$P(u_{ij} > 3\sigma^{-1} x)$	$h_{ij,ij}$	$P(u_{ij} > 3\sigma^{-1} x)$	$h_{ij,ij}$	$P(u_{ij} > 3\sigma^{-1} x)$	$h_{ij,ij}$	$P(u_{ij} > 3\sigma^{-1} x)$	$h_{ij,ij}$	$P(u_{ij} > 3\sigma^{-1} x)$
0.107	2.08×10^{-22}	0.103	6.90×10^{-9}	0.120	1.09×10^{-8}	0.105	8.67×10^{-20}	0.215	3.60×10^{-10}
0.102	1.24×10^{-17}	0.122	4.05×10^{-3}	0.114	1.40×10^{-14}	0.148	7.22×10^{-13}	0.115	1.56×10^{-14}
0.102	3.28×10^{-19}	0.101	3.05×10^{-10}	0.145	7.06×10^{-7}	0.114	4.93×10^{-15}	0.101	9.45×10^{-15}
0.120	2.23×10^{-17}	0.101	1.34×10^{-14}	0.100	0.203	0.120	1.87×10^{-8}	0.102	6.09×10^{-6}
0.138	1.09×10^{-14}	0.106	6.70×10^{-9}	0.120	1.24×10^{-15}	0.136	4.14×10^{-16}	0.133	1.18×10^{-13}
0.127	1.08×10^{-14}	0.106	1.38×10^{-12}	0.161	1.80×10^{-7}	0.103	1.39×10^{-9}	0.109	2.17×10^{-10}
0.117	1.02×10^{-6}	0.100	3.64×10^{-20}	0.103	2.23×10^{-18}	0.120	3.18×10^{-13}	0.122	6.89×10^{-6}
0.100	1.02×10^{-15}	0.103	1.18×10^{-20}	0.100	4.41×10^{-10}	0.115	1.19×10^{-11}	0.158	2.70×10^{-4}
0.101	1.56×10^{-9}	0.165	4.76×10^{-4}	0.119	6.98×10^{-9}	0.132	1.93×10^{-16}	0.100	1.64×10^{-10}
0.100	2.23×10^{-18}	0.101	1.45×10^{-2}	0.130	3.06×10^{-10}	0.165	7.57×10^{-10}	0.128	1.64×10^{-5}

The posterior probability that u_{43} is more than three standard deviations away from zero is 0.203. Observation 43 (the fourth observation in the third trial) might therefore be considered to be an outlier. It is however, not very influential because its leverage is quite small, only about 0.100. Since its influence on the estimated regression coefficients is only moderate, there is not much reason for leaving it out of the data set. Observation 15 is associated with the largest leverage, $h_{15,15} = 0.215$ and there is therefore uncertainty about u_{15} .

Using equation (A.1.5) the 95% highest posterior density (hpd) intervals for u_{ij} were calculated and are given in Van der Merwe and Van der Merwe (1990).

A.2 THE REALIZED ERROR TERMS FOR CHECKING MODELS (3.1.1)-(3.1.5)

From equations (3.1.1)-(3.1.4) it follows that the marginal distribution of x_{ij} is normal with mean

$$(A.2.1) \quad E(x_{ij} | \mu, a, \sigma_2^2) = \mu$$

and variance

$$(A.2.2) \quad Var(x_{ij} | a, \sigma_2^2) = (a+1)\sigma_2^2 = \frac{1+b(n-1)}{nb}\sigma_2^2$$

where

$$b = \frac{1}{1+an}.$$

A given observation x_{ij} satisfies the equation

$$x_{ij} = \zeta_j + \tilde{\varepsilon}_{ij} (i = 1, \dots, n; j = 1, \dots, p)$$

where $\tilde{\varepsilon}_{ij}$ is a realized error term. From the posterior distribution of $\underline{\theta}_2 = (\zeta_1 \dots \zeta_p)'$ conditional on σ_2 and b it follows that $\tilde{\varepsilon}_{ij}$ is normally distributed with mean

$$(A.2.3) \quad E(\tilde{\varepsilon}_{ij} | X, \sigma_2, b) = \hat{\varepsilon}_{ij} = (x_{ij} - \bar{x}_j) + b(\bar{x}_j - \bar{\bar{x}})$$

and variance

$$(A.2.4) \quad Var(\tilde{\varepsilon}_{ij} | X, \sigma_2, b) = \frac{\sigma_2^2}{n} \left\{ 1 - b \left(1 - \frac{1}{p} \right) \right\}.$$

The probability

$$\begin{aligned} & P(|\tilde{\varepsilon}_{ij}| > \tilde{k} \left[\frac{1+b(n-1)}{nb} \right]^{\frac{1}{2}} \sigma_2 | X) \\ &= \int_0^1 \int_0^\infty P(|\tilde{\varepsilon}_{ij}| > \tilde{k} \left[\frac{1+b(n-1)}{nb} \right]^{\frac{1}{2}} \sigma_2 | X, \sigma_2, b) p(\sigma_2 | X, b) f(b|c) d\sigma_2 db \end{aligned}$$

can therefore be defined as the probability that the ij -th observation will be an outlier.

The density function $f(b|c)$ is defined in equation (3.1.15) and

$$A.2.5 \quad p(\sigma_2 | X, b) \propto \sigma_2^{-(p+v_2)} \exp \left\{ -\frac{1}{2\sigma_2^2} (bS + W) \right\}.$$

For $\tilde{k} = 3$, these posterior probabilities are given in Table A.2.1.

Table A.2.1

The Posterior Probabilities of "Feed Consumption Data" being Outliers

$$P(|\tilde{\varepsilon}_{ij}| > \tilde{k} \left[\frac{1+b(n-1)}{nb} \right]^{\frac{1}{2}} \sigma_2 | X) \quad (\tilde{k} = 3)$$

Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
8.16×10^{-42}	6.63×10^{-42}	9.89×10^{-19}	1.99×10^{-44}	3.46×10^{-2}
1.48×10^{-24}	2.85×10^{-16}	1.11×10^{-16}	2.78×10^{-8}	2.48×10^{-14}
2.96×10^{-37}	5.76×10^{-31}	1.57×10^{-11}	8.58×10^{-18}	1.52×10^{-41}
4.31×10^{-20}	5.76×10^{-31}	1.29×10^{-34}	6.17×10^{-15}	1.76×10^{-28}
4.12×10^{-14}	1.46×10^{-44}	1.63×10^{-14}	2.79×10^{-13}	1.75×10^{-15}
3.30×10^{-10}	1.13×10^{-24}	5.78×10^{-9}	1.91×10^{-42}	6.12×10^{-26}
5.49×10^{-8}	1.87×10^{-32}	9.07×10^{-25}	6.17×10^{-15}	1.27×10^{-11}
1.81×10^{-28}	5.60×10^{-28}	3.00×10^{-34}	1.71×10^{-20}	2.83×10^{-10}
4.11×10^{-26}	4.58×10^{-7}	4.08×10^{-15}	2.35×10^{-14}	1.72×10^{-37}
4.47×10^{-28}	3.21×10^{-30}	1.76×10^{-11}	3.29×10^{-8}	6.11×10^{-17}

From the table it is clear that except for observation 15 it is unlikely that there will be outliers among the feed consumption data. The posterior probability of 0.0346 however, shows that there is a slight possibility that observation 15 (the first feed consumption value in the fifth trail) can be considered an outlier.

By using numerical integration the 95% posterior credibility interval for $\tilde{\varepsilon}_{15}$ is found to be 105.0 – 176.0. For further details, see Van der Merwe and Van der Merwe (1990).

ANOTHER LINEAR DIAGONALS-PARAMETER SYMMETRY MODEL FOR SQUARE CONTINGENCY TABLES WITH ORDERED CATEGORIES

Sadao Department of Information Sciences, Faculty of Science & Technology, Science University of Tokyo,
Tomizawa: Noda City, Chiba 278 Japan

Key words: Conditional symmetry; Extended quasi-symmetry; Odds; 2-ratios-parameter symmetry.

Summary: For square contingency tables with ordered categories, Agresti (1983) considered the linear diagonals-parameter symmetry (LDPS) model. This paper proposes another LDPS model in which the positive (negative) log odds parameters have a decreasing (increasing) linear pattern of the distance from the diagonal. The proposed model is a special case of Tomizawa's (1987) 2-ratios-parameter symmetry model, and also a simple version of Tomizawa's (1984, 1987) extended quasi-symmetry model. The model is applied to the British father-son occupational mobility data.

1. INTRODUCTION

For the $R \times R$ square contingency table with ordered categories, let p_{ij} denote the probability that an observation will fall in the cell in row i and column j ($i = 1, 2, \dots, R; j = 1, 2, \dots, R$). Goodman (1979) considered the diagonals-parameter symmetry (DPS) model, defined by

$$p_{ij} = \begin{cases} \delta_{j-i}\psi_{ij} & \text{for } i < j, \\ \psi_{ij} & \text{for } i \geq j, \end{cases}$$